Engineered enterprise banking applications for BRAC Bank

What it means

You developed banking applications used by customers through:

- Internet Banking (Web)
- Mobile Banking App
- Internal Banking Systems

These applications communicated with middleware and core banking systems to process transactions.

Interview Explanation

I worked on digital banking applications for BRAC Bank. My responsibility was to develop backend services that handled customer transactions, account information, fund transfers, bill payments, and communication with middleware and core banking systems. I ensured secure and reliable transaction processing across web and mobile channels.

2. Implemented SOAP-based integrations by consuming WSDL and XSD contracts

Important Terms

SOAP

A protocol used for communication between applications using XML.

WSDL (Web Service Description Language)

A contract that tells:

- What service is available
- What operations can be called
- Request and response structure

XSD (XML Schema Definition)

Defines the XML format and validation rules.

What you did

- Received WSDL from middleware team
- Generated client classes
- Sent SOAP requests
- Validated XML against XSD
- Transformed data between systems

Interview Explanation

I integrated our application with middleware systems using SOAP services. The middleware team provided WSDL and XSD files. I generated Java client classes, mapped request/response objects, validated XML messages, and handled data transformation according to banking standards.

3. Developed and exposed REST APIs for mobile applications and SOAP services for web applications

REST API

Used mainly by mobile applications.

Example:

GET /accounts/12345

Returns:

```
{
  "accountNumber":"12345",
  "balance":5000
}
```

SOAP Service

Used by legacy web applications and middleware.

What you did

- Developed REST APIs in Spring Boot
- Developed SOAP services
- Implemented authentication and authorization
- Applied business rules consistently

Interview Explanation

I developed REST APIs for mobile banking applications and SOAP services for web channels. Both services shared common business logic, validation rules, and security controls to ensure consistency across channels.

4. Built digital payment modules including IDTP and QR scan-based transactions

IDTP

Interoperable Digital Transaction Platform.

Allows money transfers between different banks and payment systems.

QR Payment Flow

Add More

```
.graphable-function-v2-jsxgraph {
  position: relative;
  z-index: 1;
  width: 100%;
  height: 100%;
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```

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  background: transparent;
}

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  outline: none;
}

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  margin-top: -4px;
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  border-radius: 9999px;
  pointer-events: none;
  z-index: 2;
}

.graphable-function-v2-jsxgraph-tooltip {
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  z-index: 3;
  overflow: visible;
  pointer-events: none;
}

.graphable-function-v2-jsxgraph-reset {
  position: absolute;
  left: 12px;
  bottom: 12px;
  display: inline-flex;
  align-items: center;
  justify-content: center;
```

```
    z-index: 2;
    width: 32px;
    height: 32px;
    border: 1px solid var(--border-light);
    border-radius: 9999px;
    background: var(--bg-primary);
    color: var(--text-primary);
    box-shadow: 0 2px 8px rgb(0 0 0 / 8%);
    padding: 0;
    transition: opacity 160ms ease, transform 160ms ease;
  }
246-2-4-61234-1-2-3-4-5-60,0
```

Give feedback

(Widget not actually needed here conceptually; ignore in interview.)

Actual flow:

Customer scans QR →

Mobile App →

Backend Service →

Payment Gateway →

Banking System →

Transaction Success

Polling Mechanism

After payment request:

```
while(status.equals("PENDING")){
    checkTransactionStatus();
}
```

System repeatedly checks transaction status until completed.

Interview Explanation

I worked on digital payment modules such as IDTP transfers and QR-code-based payments. Since transaction processing was asynchronous, we implemented polling mechanisms to continuously check transaction status and update customers once the payment was completed successfully.

5. Designed and optimized performance-critical Oracle SQL queries

What it means

You improved database performance.

Complex Joins

Example:

```
SELECT c.customer_name,
        a.account_no,
        t.amount
FROM customer c
JOIN account a
ON c.customer_id=a.customer_id
JOIN transaction t
ON a.account_id=t.account_id;
```

Optimization Techniques

- Created indexes
- Reduced full table scans
- Optimized joins
- Used execution plans
- Improved query structure

Interview Explanation

I optimized Oracle SQL queries used in transaction processing and reporting modules. By analyzing execution plans, improving joins, and utilizing indexes, I significantly reduced query execution time and improved application performance.

6. Managed production support and Jenkins CI/CD pipelines

Production Support

When issues occur in production:

- Analyze logs
- Identify root cause
- Fix defects
- Deploy patches

RCA (Root Cause Analysis)

Example:

Issue:

Fund transfer failed.

Root Cause:

Database connection pool exhausted.

Resolution:

Increased pool size and optimized connection handling.

Jenkins CI/CD

Flow:

Developer Commit

↓

Git

↓

Jenkins Build

↓

Unit Testing

↓

Deployment

↓

Production

Zero-Downtime Deployment

Users continue using the application while new version is deployed.

IBM WebSphere

Enterprise application server for Java applications.

AWS EC2

Cloud virtual machines where applications are hosted.

Interview Explanation

I managed production incidents, analyzed logs, prepared RCA reports, and coordinated fixes. I also maintained Jenkins CI/CD pipelines for automated build and deployment processes, ensuring zero-downtime releases on IBM WebSphere and AWS EC2 environments.

7. Adopted GitHub Copilot into the development workflow

GitHub Copilot

AI-powered coding assistant.

Used for:

- Code generation
- Unit tests
- Boilerplate code
- Refactoring suggestions

Benefits

- Faster development
- Better productivity
- Reduced repetitive coding

Interview Explanation

I integrated GitHub Copilot into our development process to accelerate coding activities,

generate boilerplate code, create unit tests, and improve developer productivity while maintaining coding standards through code reviews.

Overall Role Summary (Interview Answer)

I worked as a Backend Java Developer in the digital banking domain for BRAC Bank. My responsibilities included developing REST and SOAP services, integrating with middleware and core banking systems, implementing digital payment solutions such as QR and IDTP transactions, optimizing Oracle database queries, handling production support, managing Jenkins CI/CD deployments, and ensuring secure, reliable transaction processing across web and mobile banking platforms.